

Macro Copilot

A deterministic analytics engine for discretionary macro research.

TOOL CATALOGUE & PRODUCT OVERVIEW

Rates & FX Agents, MVP scope.

April 2026. This document describes the architecture, design principles, and tool inventory of the Macro Copilot rates and FX agents. It is intended as a reference for advisors, prospective partners, and internal stakeholders. The product is under active development; screenshots and tool counts reflect the current MVP state.

1. Vision & Problem Statement

The information problem in macro research

Macro research teams face a growing information problem. Portfolio managers and analysts must process huge volumes of material every day: sell-side research, central bank speeches, economic releases, academic work, internal research, broker commentary. Traditionally, answering a broad macro question such as *"What is happening in Australia and are there trades to do?"* could take several hours of finding the relevant sources, reading them, extracting the key points, and synthesising them into a concise investment-oriented summary.

The rise of generative AI changes this workflow. Large language models can help consume, filter, summarise, and aggregate large volumes of research much faster than a human analyst. In a recent JP Morgan Market Matters discussion with Chris Pullman, Head of Macro Research and Chief Economist at Balyasny Asset Management, Pullman explains that a process which previously took around seven hours can now be reduced to a few minutes when supported by a robust AI and data infrastructure.

However, the main insight is that the value does not come from simply "adding ChatGPT" on top of research documents. The competitive advantage comes from building a robust data and modelling pipeline based on software-engineering principles, capable of handling multidimensional data, point-in-time economic datasets, research libraries, and model outputs in a scalable and reliable way. The LLM is the interface; the engine underneath is what matters.

The specific bottleneck on a discretionary macro pod

Discretionary macro trading desks at multi-manager platforms operate in pods of five to six people: a portfolio manager, a few analysts or researchers, occasionally a junior quant. These teams sit atop institutional-grade data (Bloomberg, internal time-series databases, broker research) and employ significant talent to analyse it. The data quality is high. The analytical skill is high. What is structurally lacking is **speed to insight**.

The bottleneck is not data or talent. It is the repetitive, manual workflow that sits between a PM's question and a quantitative answer. *"Where are 2s10s relative to the last year? What is the market pricing for the next three Fed meetings? Scan me for any sovereign yield at a 2-sigma extreme."* These are answerable questions with deterministic, auditable computations, but answering them today requires an analyst to open a terminal, pull data, write or run a script, format the output, and relay it back. Each pod typically rebuilds the same scripts and dashboards from scratch as people rotate or mandates change.

What Macro Copilot delivers

Macro Copilot is a productised solution to this problem. It integrates a multi-agent LLM architecture with a deterministic Python analytics engine and an autonomous data layer that connects to the fund's existing data sources. The LLM layer handles natural-language understanding, workflow orchestration, and result interpretation. The compute layer handles all quantitative work. Every number traces to a named tool. Every tool traces to a data source and a timestamp. **The LLM does zero math.** The product augments the PM's research workflow, compressing typical research latency from minutes or hours down to seconds, while preserving full auditability of every figure surfaced.

2. Multi-Agent Architecture

The system is organised into four layers. Layer 0 is the user-facing PM Orchestrator that receives natural-language queries and routes them. Layer 1 consists of domain agents (Rates, FX, Credit, Policy and Events), each with isolated tool access and a domain-specific system prompt. Layer 2 breaks each domain agent into instrument-specific sub-agents. Each sub-agent dispatches to deterministic Python tools via MCP (Model Context Protocol) servers. All tool calls, inputs, outputs, and data timestamps are logged and surfaced in the UI.

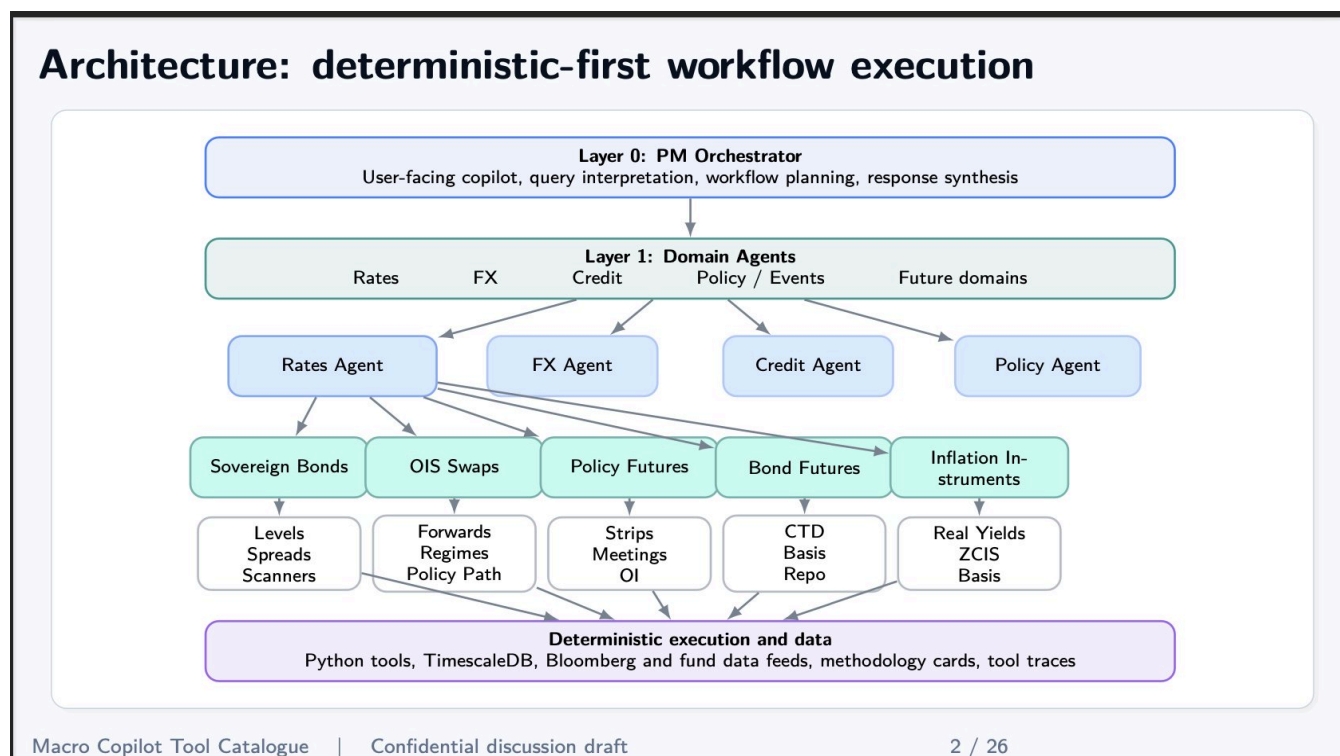


Figure 1. Layered multi-agent architecture. Layer 0: PM Orchestrator (user-facing copilot, query interpretation, workflow planning, response synthesis). Layer 1: domain agents (Rates, FX, Credit, Policy / Events, future domains). The Rates Agent is shown expanded into its five instrument sub-agents, each with representative tool categories. All execution flows down to the deterministic layer: Python tools, TimescaleDB, Bloomberg and fund data feeds, methodology cards, and tool traces.

Why this architecture

Three properties matter. First, **domain isolation**: the Rates agent cannot call FX tools and vice versa, so tool selection is unambiguous and domain-specific reasoning improves. Second, **deterministic compute**: the LLM selects which tool to call and how to interpret the result, but never computes a number itself; all math is delegated to versioned Python tools backed by TimescaleDB. Third, **auditability**: every figure on the screen has an inspection layer showing the tool, parameters, data source, and timestamp.

3. Design Principles

Four design principles govern every tool, workflow, and UI decision. These are not aspirations but hard constraints, and they are what differentiate the product from a general-purpose LLM wrapper.

01

Deterministic-first.

The LLM does zero math. Every quantitative output, including yields, spreads, z-scores, forward rates, carry estimates, and regime labels, is computed by a named, versioned Python tool operating on timestamped data from TimescaleDB. The LLM understands the PM's intent, selects the right tools, passes the right parameters, and translates the result into natural language. If the tool returns 76.2bp, the copilot reports 76.2bp. It does not round, interpolate, or hallucinate.

02

Full traceability.

Every number in the UI traces to (1) the tool that produced it, (2) the exact inputs and parameters, (3) the data source and timestamp of the underlying data, and (4) the methodology and assumptions used. The PM can click any result to open an inspection layer showing this complete chain. There are no black boxes.

03

Bucket discipline.

Every tool is classified into one of three buckets, with a fourth tier for aspirational tools. **Bucket 1A**: strict deterministic, fixed defaults, one correct answer given inputs. **Bucket 1B**: parameterised deterministic, PM-overridable assumptions (lookback windows, models, thresholds, conventions). **Bucket 2**: stochastic, latent-factor, or probabilistic models (PCA, HMM, GARCH, term-premium decompositions) where output is model-dependent and must be labelled as such.

04

Domain isolation.

Each domain agent has hard-wired access only to its own tools and data. The Rates agent cannot call FX tools, and vice versa. Cross-domain queries (for example, FX-hedged yield pickup) are orchestrated at Layer 0, which calls both agents and composes the result. This eliminates tool-selection ambiguity and ensures each agent is an expert in its domain.

Empirical validation. On the current rates agent MVP, a 30-prompt validation gauntlet achieved 29 of 29 correct tool routings, 29 of 29 correct tool selections, and 59 of 59 mathematically accurate outputs against SQL ground-truth values.

4. User Interface

The product ships as a web application with four primary surfaces, each designed for a different mode of PM interaction. The interface enforces the traceability principle: every number is clickable, every tool call is inspectable, and the underlying computation is always one click away.

Daily Monitor

Home screen. Morning snapshot covering top z-score extremes, curve regime classifications, PCA factor summaries, cross-asset market movers, and macro highlights. Updated in real time.

Copilot

Core product. Natural-language chat where the PM asks questions and receives deterministic, tool-backed answers with inline charts, tables, and a tool-chain trace. Every response shows tool count, execution time, and a link to the full audit trail.

Workspace

Flexible analytical canvas. Open any instrument, spread, or model result in a dedicated panel with interactive controls. Lookback windows, tenors, and overlays are adjustable; the copilot stays in context for follow-up questions.

Tool Library

Browsable catalogue of every tool, with bucket label, input/output schema, methodology card, and example usage. Each tool exposes *What it does*, *How it works*, *Assumptions and Controls*, and *Interpretation* tabs.

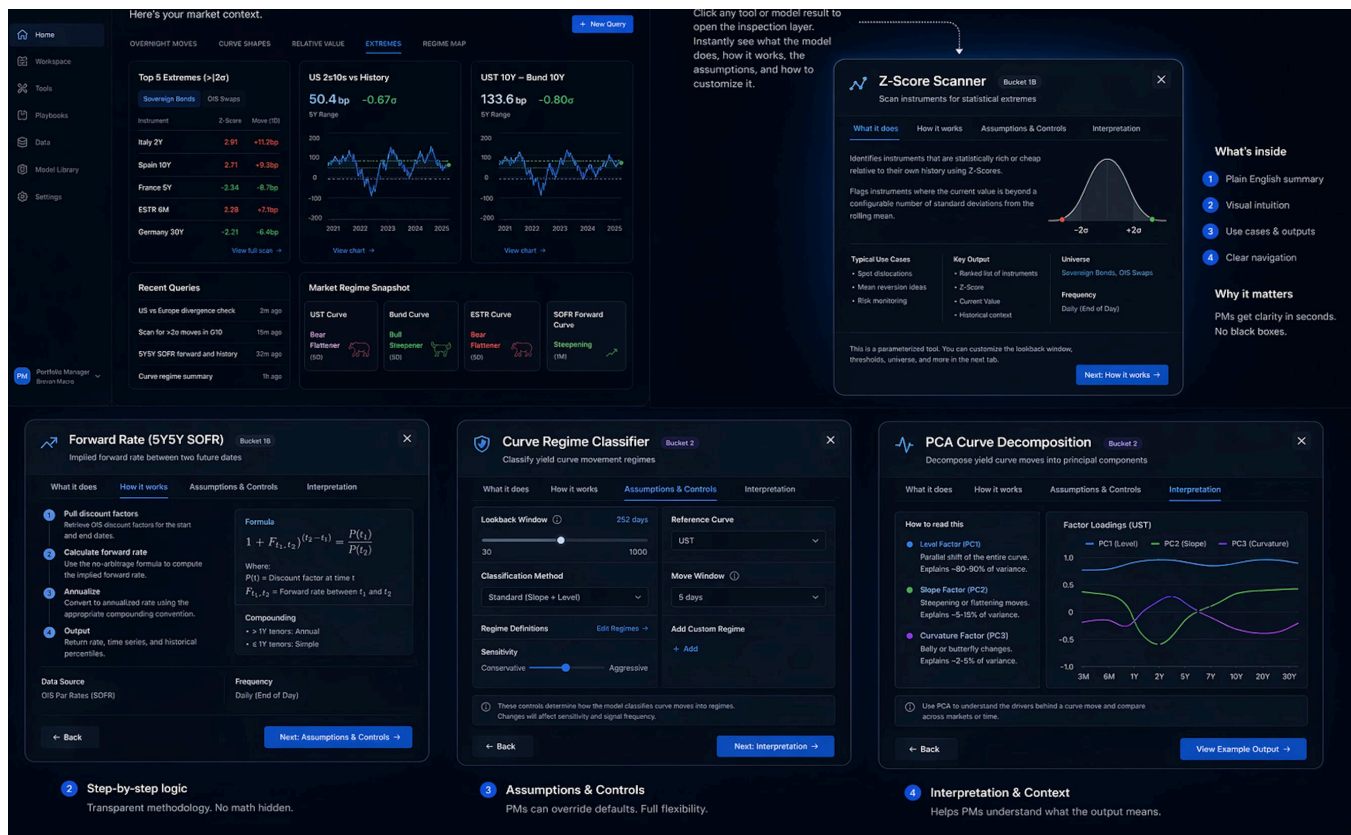


Figure 2. Representative product design. The Daily Monitor (top left) shows extremes, curve shapes, and regime snapshots. Inspecting any tool result (top right) opens a four-tab panel with plain-English summary, methodology, assumptions and controls, and interpretation. Bottom row: individual tool cards for Forward Rate (1B), Curve Regime Classifier (2), and PCA Curve Decomposition (2).

4.1 Current MVP, working screenshots

The screenshots below are taken from the working v0.1 MVP, running locally against a TimescaleDB instance with daily Bloomberg-sourced data ingested via xbbg. The UI is functional but pre-design polish.

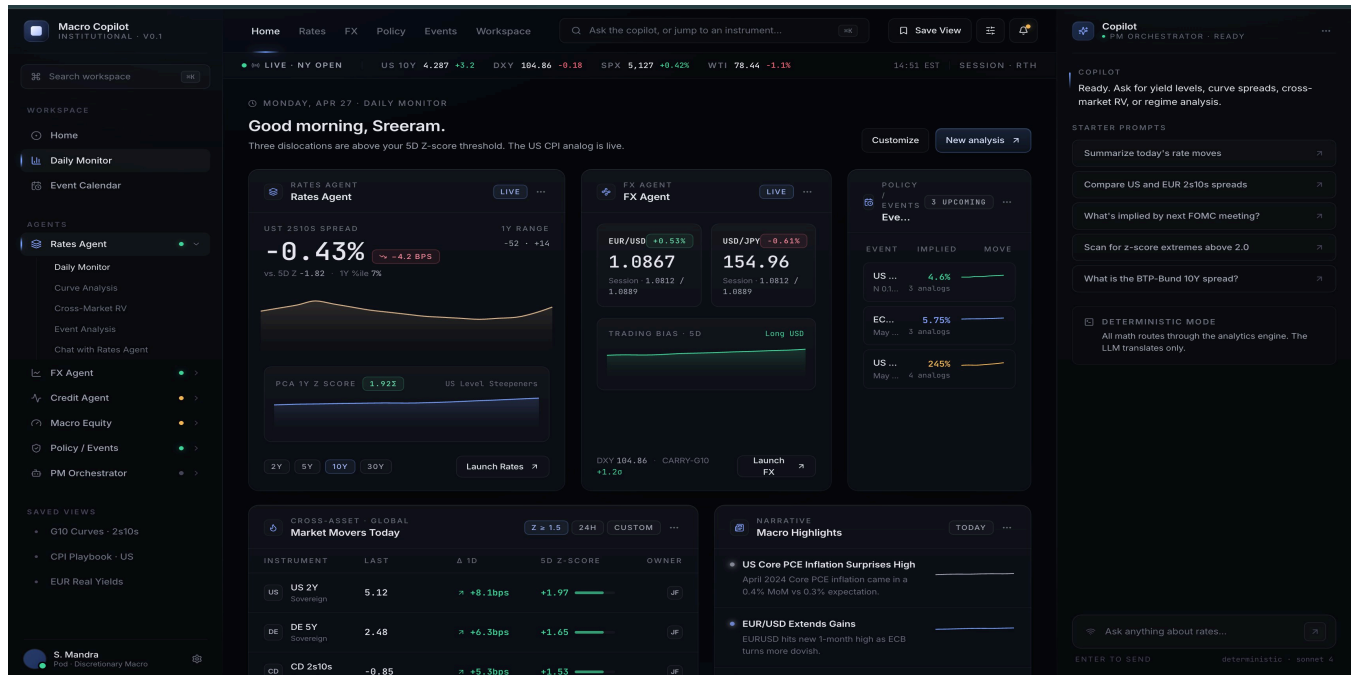


Figure 3. **Daily Monitor.** Rates agent summary (UST 2s10s, PCA z-score), FX agent summary (EUR/USD, USD/JPY with session ranges and trading bias), upcoming policy events with implied moves and analog matches, cross-asset market movers, and macro narrative highlights. The right rail hosts the persistent copilot.

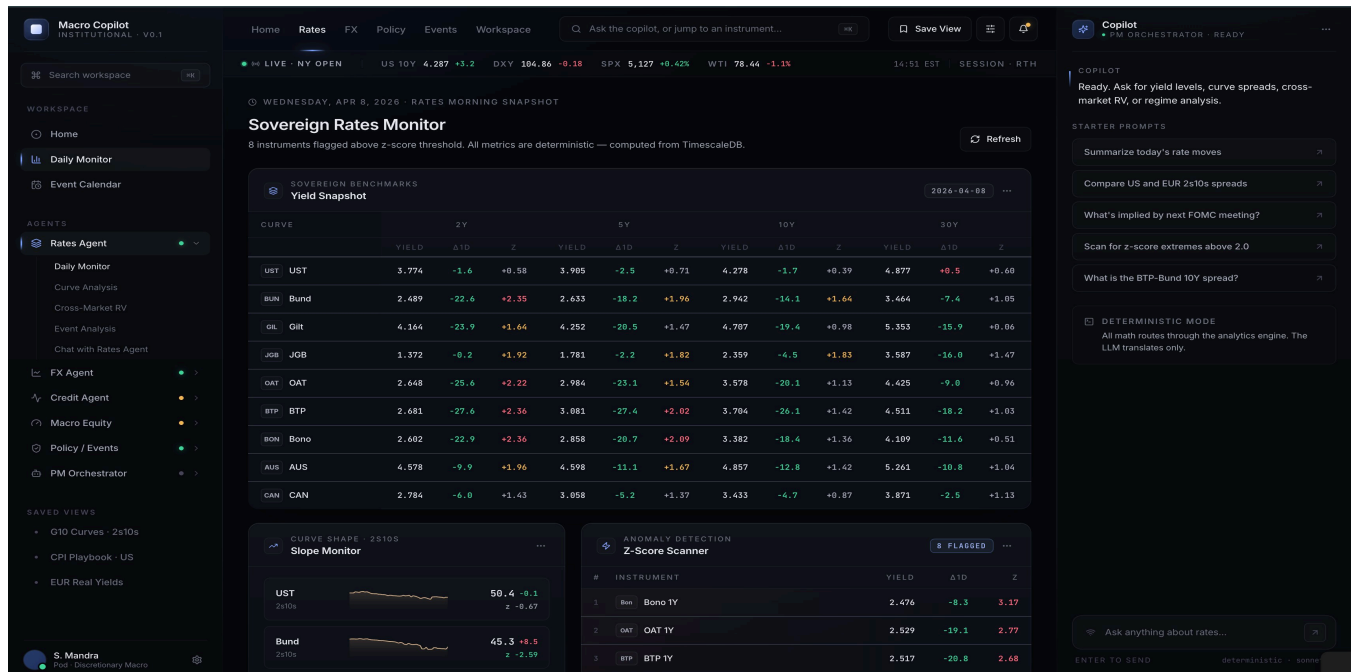


Figure 4. **Sovereign Rates Monitor.** Yield snapshot across nine curves (UST, Bund, Gilt, JGB, OAT, BTP, Bono, AUS, CAN) at 2Y, 5Y, 10Y, and 30Y with daily changes and 252d z-scores. Below: the slope monitor surfaces curve shape per country, and the z-score scanner ranks instruments above the user's threshold.

4.2 Copilot in action

The screenshot below shows the copilot executing a multi-tool deterministic workflow in response to the natural-language prompt "I need a quick global macro matrix." The system pulls 10-year US and German yield levels, 5-day 2s10s curve regimes for both markets, the US 10Y versus Germany 10Y cross-market spread, and highlights z-scores above 1.5. Total latency: 16.0 seconds.

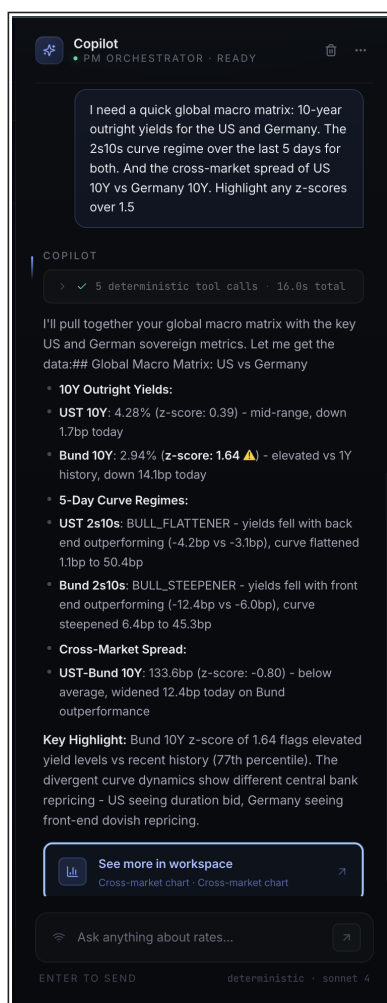


Figure 5. Copilot composing a multi-tool global macro matrix. The prompt asks for US and Germany 10-year yields, 5-day 2s10s regimes, the UST-Bund 10Y cross-market spread, and z-score highlights. The response shows 5 deterministic tool calls, then synthesises the result into a PM-readable narrative while preserving a link back to the workspace for inspection.

What this shows. The copilot is not summarising free-text research. It is calling deterministic Python tools, receiving structured numerical outputs, and translating those outputs into a PM-readable narrative. The visible tool trace shows that five tools were executed. The "See more in workspace" link returns the full tool output and charts for inspection, preserving traceability from response to model, tool, and source data.

5. Tool Taxonomy

Every tool in the system carries a bucket label. The classification determines how output is presented, what controls the PM has, and how uncertainty is communicated in the natural-language response.

BUCKET	CHARACTER	WHAT IT MEANS	PM CONTROLS	EXAMPLE TOOLS
1A	Strict deterministic, fixed defaults	Bloomberg-replica tier. One correct answer given inputs. No assumptions to override. Used everywhere a single canonical figure is needed.	None. Output is fixed given the query.	yield_levels curve_spread scanner
1B	Parameterised deterministic	Deterministic given parameters, but the parameters encode assumptions a PM may wish to change. Same inputs always return the same answer. The methodology card displays every chosen parameter.	Lookback window, model choice, thresholds, conventions, financing.	curve_fitter carry_and_roll per_meeting_pricing
2	Stochastic, model-dependent	Output depends on a fitted statistical model. Different specifications produce different answers. Always labelled model-dependent in the natural-language response.	Model spec, number of states, feature set, lookback.	pca_yield_curve hmm_regime term_premium_acm
Asp	Aspirational	Documented for roadmap completeness but currently data-blocked or further out in the build sequence.	n/a	repo_specialness flow_attribution

Why bucket labels appear in every response. When a PM sees a yield level of 4.287% on screen, the 1A label tells them the figure is the canonical TimescaleDB observation, not a model output. When they see a term-premium decomposition, the 2 label tells them the number is model-dependent and would differ under an alternative specification. The label is a contract between the product and the user about how much to trust each figure and where the relevant uncertainty sits.

Rates Universe

Six instrument domains. Seventy-plus tools across three buckets.

Nine sovereign curves · six OIS curves · nineteen bond futures contracts
twenty-four inflation-indexed bonds · twenty-one inflation swaps · twenty-four policy futures.

5.1 Sovereign Bonds

9 countries: UST, Bund, Gilt, JGB, OAT, BTP, CAN, AUS, Bono. Four tenors per curve (2Y, 5Y, 10Y, 30Y).

Sovereign bond yields are the foundational price in macro. They encode monetary policy expectations, term premium, fiscal and supply dynamics, and global risk appetite simultaneously. The sovereign module provides level, curve-shape, cross-market, and regime analytics across the G10. Breakeven inflation, the spread between nominal yields here and real yields in the inflation-linked bond module, is a derived cross-module metric.

Tool Inventory

1A **yield_levels**

Returns the latest yield for any (country, tenor) pair along with period changes (1d, 5d, 22d), 252d high and low, percentile rank, and z-score. Foundational primitive used by every directional and screening workflow as the entry point for yield-level analysis.

1A **curve_spread**

Computes the yield differential between two tenors on a single curve and returns the time series, current value, period changes, and 252d z-score. UST 2s10s, Bund 5s30s, and similar curve-shape metrics route through this tool.

1A **cross_market_spread**

Returns the differential between two (country, tenor) pairs at matched dates with full statistical context. UST minus Bund 10Y, BTP minus Bund 10Y, and similar cross-country spreads are the primary use case.

1A **butterfly**

Computes the curvature spread (2 times belly minus the two wings) at a chosen country and tenor triplet, with default 50-50 wing weighting. Returns current value, z-score, and per-wing decomposition. Used to surface belly richness or cheapness independent of overall curve direction.

1A **curve_regime**

Classifies the curve move over a chosen lookback window as bull-steepener, bear-flattener, bull-flattener, bear-steepener, parallel shift, or twist using fixed default thresholds on front-end and long-end directional changes. Used in morning briefing and regime workflows.

1A **scanner**

Iterates yield_levels across all (country, tenor) combinations and returns those exceeding a z-score threshold (default 2.0), ranked by magnitude. The workhorse tool for surfacing dislocation across the global sovereign universe in one call.

1B **curve_fitter**

Fits Nelson-Siegel, Svensson, or cubic-spline term-structure models to the yield curve at a date and returns fitted parameters, fitted curve at all tenors, and per-bond rich/cheap residuals. The PM may select the model class. (Nelson and Siegel 1987; Svensson 1994)

1B **carry_and_roll**

Computes expected return from carry plus roll-down for a chosen instrument and holding horizon (1M, 3M, 6M). Decomposes total return into coupon income, roll-down on a chosen curve interpolation, and pull-to-par. Financing assumption is PM-overridable (custom repo, constant rate, or implied from OIS).

1B **asset_swap_spread**

Computes par-ASW, I-spread, or Z-spread against a chosen swap curve. Bridges the sovereign and OIS modules. Convention selection matters: par-ASW for near-par RV, I-spread for quick comparisons, Z-spread for credit-adjacent or off-par bonds. (Tuckman Ch. 17)

1B beta_adjusted_spread

Runs a rolling OLS of one yield series on another and returns rolling beta, residuals, and residual z-score. Useful in cross-market RV where raw spreads are misleading because the two markets do not share equal sensitivity to common drivers.

1B historical_replay

Replays a historical window of daily yield and curve changes through a user-supplied position spec and returns the P&L path, drawdown, vol, worst day, and decomposition by instrument. The 2013 taper tantrum, 2020 COVID, and 2022 Gilt LDI windows are preconfigured stress cases.

2 pca_yield_curve

Computes principal components of yield changes across a chosen tenor universe and lookback, returning level, slope, and curvature loadings, factor time series, and variance explained. Foundation for factor attribution and most quantitative rates work. (Litterman and Scheinkman 1991)

2 hmm_curve_regime

Fits a Gaussian Hidden Markov Model on a feature vector (yield change, slope change, MOVE level) with a chosen number of states. Returns state probabilities, transition matrix, current state, and per-state feature distributions. Output is regime-classification model-dependent. (Hamilton 1989; Rabiner 1989)

2 term_premium_acm

Estimates the Adrian-Crump-Moench five-factor affine term-structure model and decomposes yields into expected short-rate and term-premium components. Used to answer whether a yield move is policy-expectation-driven or term-premium-driven. Different specifications produce materially different premium series. (Adrian, Crump and Moench 2013)

2 garch_yield_vol

Fits a GARCH(1,1) model to a yield series and returns the conditional volatility forecast and vol-of-vol. Used in vol-regime workflows and volatility-targeted position sizing. (Bollerslev 1986)

5.2 Overnight Index Swaps (OIS)

6 currencies: USD SOFR, EUR ESTR, GBP SONIA, JPY TONA, AUD AONIA, CAD CORRA. Thirteen tenors each (1W to 30Y).

Overnight index swaps are the cleanest measure of risk-neutral monetary policy expectations in each currency. Unlike sovereign yields, OIS rates are free of fiscal supply, credit, and liquidity-convenience effects, making the OIS curve the purest expression of the expected policy path.

The front end (sub-2Y) prices the trajectory of central bank rate decisions; the back end (5Y and beyond) prices the terminal rate and OIS term premium. Per-meeting pricing and forward rates are the primary derived analytics PMs ask for, and the cross-CB OIS spread (SOFR minus ESTR at 2Y) is the canonical measure of Fed-versus-ECB cycle divergence.

Tool Inventory

1A rate_level

Returns the OIS rate for any (curve_family, tenor) with period changes (1d, 5d, 22d), 252d high, low, percentile, and z-score. The OIS analogue of yield_levels and the entry point for every front-end workflow.

1A curve_spread

OIS tenor differential at chosen curve_family. The cleanest curve-shape measure on a desk: no supply or credit noise, only the term structure of expected policy.

1A forward_rate

Implied forward rate computed from two OIS pillar rates using annual compounding for windows over 1Y. 1Y1Y, 2Y1Y, and 5Y5Y are the canonical outputs. Used for terminal-rate proxies and forward-trade workflows.

1A cross_market_spread

OIS rate differential between two (curve_family, tenor) pairs. SOFR minus ESTR at 2Y is the canonical Fed-versus-ECB front-end-cycle divergence metric.

1A scanner

Iterates rate_level across the full 6-curve, 13-tenor OIS grid and returns z-score extremes. Used in morning briefing to surface where OIS markets are showing statistical stress.

1B per_meeting_pricing

Decomposes the front-end OIS strip into per-meeting risk-neutral implied policy changes and probabilities, given the central bank's meeting calendar and explicit adjustments (turn-of-year, terminal-rate assumption, stub-period handling). **The single most-asked metric on a macro desk.** Risk-neutral implied, not real-world expected; the wedge is surfaced as an explicit assumption rather than absorbed silently.

1B policy_path_since_event

Computes the change in per-meeting implied path between two reference dates (e.g., pre-CPI vs post-CPI, pre-FOMC vs post-FOMC). Returns the path-difference profile and total terminal-rate move. Typically the second-most-asked CB-pricing metric after raw meeting pricing.

1B zero_curve_bootstrap

Strips the OIS pillar grid into a zero curve and discount factors using a chosen interpolation method (linear, log-linear, monotone cubic spline). Foundational for downstream tools: ASW, exact forwards, parametric scenarios all depend on this.

1B curve_fitter_ois

Nelson-Siegel, Svensson, or cubic-spline fit on the OIS curve. Returns fitted parameters, fitted curve, and per-pillar residuals. Used to identify mispriced pillar tenors and as a smoothing layer for downstream tools.

1B beta_adjusted_ois_spread

Rolling OLS of one OIS series on another, returning rolling beta, residual, and residual z-score. Cross-CB OIS spreads are not 1:1; the Fed and ECB respond at different magnitudes to common shocks, and the beta-adjusted spread captures the divergence signal cleanly.

1B historical_replay

Replays an OIS-rate path through a position spec across a chosen window. The 2022 Fed hiking cycle and the 2023 banking stress are preconfigured cases.

2 pca_ois_curve

PCA on OIS rate changes across a chosen tenor universe. The factor structure is particularly clean on OIS because there is no supply or credit confound. Returns level, slope, and curvature loadings, factor time series, and variance explained.

2 hmm_ois_regime

Gaussian HMM on OIS features (levels, slopes, realised vol) returning state probabilities, transition matrix, and current state. The OIS analogue is cleaner than the sovereign analogue. (Hamilton 1989)

2 term_premium_ois_acm

ACM-style affine model fit on OIS specifically, decomposing OIS rates into expected short-rate and OIS term-premium components. Cleaner than the sovereign analogue but still model-dependent.

2 multi_currency_ois_pca

Joint PCA across all six central bank OIS curves. Returns global policy-cycle factor loadings and per-CB residual exposures. The engineering challenge is calendar synchronisation across six business-day calendars.

5.3 Government Bond Futures

19 contracts, 9 countries. CME (TU, FV, TY, UXY, US, WN), Eurex (Schatz, Bobl, Bund, Buxl, BTP, OAT, Bono), ICE (Long Gilt), JPX (JGB), TMX (CGB), ASX (3Y, 10Y).

Bond futures are the primary vehicle for expressing duration and curve views in macro. They are more liquid than cash bonds, require no financing of the long, and allow rapid position adjustment. The critical analytical layer is the cheapest-to-deliver (CTD), basis, and implied-repo framework that links futures to the underlying cash bonds.

The March 2020 basis-trade unwind (an estimated \$62B in hedge-fund short reductions) and the September 2022 UK LDI crisis are the canonical stress episodes for basis monitoring and the calibration anchors for stress thresholds in the `basis_trade_monitor` tool.

Tool Inventory

1A `futures_price_level`

Returns price (in points, percent of par, or yield, depending on quoting convention), period changes, 252d high, low, percentile, and z-score for any contract. Note: ASX 3Y and 10Y futures are quoted as 100 minus yield with variable tick value, handled separately.

1A `futures_volume_oi`

Returns daily volume, open interest, and changes thereof, plus 252d z-scores on both. The primary positioning-flow primitive: a volume spike with OI build is a position-establishment signal; with OI fall is a position-unwind signal.

1A `scanner_futures`

Iterates `futures_price_level` across the full contract universe and returns z-score extremes ranked by magnitude. Companion to the sovereign scanner.

1B `ctd_identifier`

Computes the implied repo rate per deliverable bond and returns the CTD (highest implied repo) along with the full basket ranked. **Foundational**: every basis, DV01, and quality-option calculation downstream depends on this.

1B `futures_dv01`

Computes DV01 per contract from the CTD, either analytically as `DV01_CTD` divided by conversion factor, or via numerical bump (1bp shift in CTD-implied yield). Foundational for all hedging and curve-trade calculations.

1B `net_basis`

Computes net basis (gross basis minus carry) for a deliverable. Net basis is close to zero for the CTD by delivery in normal markets; meaningful deviation is a stress signal. The March 2020 episode is the calibration anchor.

1B `basis_trade_monitor`

Tracks the time series of (implied repo minus actual term repo) with 252d percentile and explicit blow-out flag if outside historical norms. Used for funding-stress monitoring.

1B `inter_commodity_spread_dv01`

DV01-neutral intra-country futures-curve spread (e.g., TU minus TY for the US futures-implied 2s10s). Uses `futures_dv01` for each leg to ensure risk neutrality.

1B `cross_country_futures_spread`

FX-adjusted, DV01-neutral cross-country futures spread. The canonical macro RV expression for cross-country rate divergence views, used in relative-value workflows. Cross-domain dependency on FX spot.

1B calendar_roll_analytics

Computes current calendar spread, fraction-rolled-versus-historical, and expected roll cost from term repo. Used for pre-roll-window planning each contract cycle.

1B stress_replay_futures

Replays preconfigured historical windows (March 2020 COVID, September 2022 LDI, December 2022 BoJ YCC band shift) through a futures position spec. Returns P&L path, drawdown, and vol.

2 pca_futures_curve

PCA on per-country bond-futures price changes. Particularly clean factor structure on the US complex (TU, FV, TY, UXY, US, WN) which spans 2Y to 30Y. (Litterman and Scheinkman 1991)

2 hmm_futures_regime

Gaussian HMM on futures features (price changes, basis, vol, positioning). Returns regime probabilities and transition matrix. (Hamilton 1989)

2 quality_option_valuation

Simulates yield-curve paths under a calibrated short-rate model (Hull-White or G2++) and values the embedded CTD-switch, end-of-month, and wildcard delivery options. Most relevant near regime transitions when CTD switching is most likely. (Hull-White 1990)

5.4 Inflation-Indexed Bonds

24 instruments, 4 countries. US TIPS (5Y / 10Y / 20Y / 30Y), UK Index-Linked Gilts (1Y to 50Y), French OATei (2Y to 15Y), Canadian RRBs (5Y to 30Y, legacy).

Inflation-indexed bonds, also called linkers, pay a fixed real coupon on an inflation-adjusted principal. The real yield is the yield-to-maturity computed on the inflation-linked cash flows; the spread between a nominal sovereign yield and the matched-tenor real yield is the breakeven inflation rate, which represents market inflation compensation for that horizon (not a pure forecast of expected inflation, due to inflation risk premium and linker liquidity premium).

Two regime breaks materially affect this universe. The UK has scheduled the alignment of RPI to CPIH from 2030 with no compensation to gilt holders, implying roughly 1 percent per annum lower indexation for post-2030 cash flows. Canada ceased new RRB issuance in November 2022, leaving the CAD linker curve as a declining-liquidity legacy market. Linkers also carry real-rate duration: TIPS lost approximately 12 percent in 2022 despite CPI running at 6.5 percent, because real yields rose sharply.

Tool Inventory

1A **real_yield_level**

Returns the real yield for any (curve_family, tenor) along with period changes, 252d high, low, percentile, and z-score. The linker analogue of yield_levels.

1A **breakeven_inflation_simple**

Joins nominal generic yield from the sovereign module and real generic yield from the linker module at matched tenor and returns breakeven level, period changes, and z-score. Output is labelled generic breakeven or inflation compensation, not a pure expected-inflation reading.

1A **forward_breakeven_simple**

Computes forward breakeven from spot breakeven points using a fixed displayed formula. The 5Y5Y forward breakeven is the Fed's preferred long-run inflation-expectations anchor, currently around 2.13 percent (April 2026).

1A **breakeven_curve_spread**

Differential between two breakeven tenors on a single country. Surfaces the term structure of inflation compensation. Front-end-high and back-end-anchored is consistent with a transitory-inflation reading; flat or inverted suggests persistent inflation.

1A **cross_country_breakeven_spread**

Raw breakeven differential between two countries at matched tenor, with explicit index-family caveat (CPI-U vs HICPxT vs RPI). Used to surface cross-country inflation-expectation divergence.

1A **scanner_linkers**

Scans real yield levels and generic breakevens across the full linker universe for z-score extremes, with liquidity and confidence flags.

1B **tips_carry_seasonal**

Projects TIPS index-ratio accrual using historical CPI-U seasonal patterns over a chosen holding horizon (1M, 3M, 6M). Returns expected carry in bps decomposed into inflation accrual, real-yield carry, and roll. Seasonal model is PM-overrideable (trailing 5Y, trailing 10Y, BLS seasonal factors, custom).

1B **carry_adjusted_breakeven**

Adjusts a static breakeven for the expected seasonal CPI carry over a chosen holding period. Answers the question: is the breakeven cheap or rich after carry?

1B breakeven_curve_fitter

Nelson-Siegel, Svensson, or spline fit on the breakeven term structure. Returns fitted curve and per-tenor residuals. Used in linker-vs-nominal rich/cheap screens.

1B uk_rpi_cpih_basis_monitor

Estimates the market-implied RPI-CPIH wedge priced into UK linkers maturing before vs after the 2030 transition. Outputs implied differential and time series with methodology card.

1B swap_breakeven_basis_simple

Computes the canonical (ZCIS minus bond-implied breakeven) basis at matched country and tenor. Surfaces linker-specific liquidity or supply-demand distortion. The primary cross-module link to the inflation swaps universe.

1B historical_replay_linkers

Replays real-yield and breakeven paths through a position spec across a window. Decomposes P&L into real-yield, breakeven-change, carry, and inflation-accrual components. The 2008 TIPS liquidity crisis and the 2020 to 2022 inflation surge are preconfigured.

2 pca_breakeven_curve

PCA on breakeven changes across tenors. Reveals whether breakeven moves are parallel (broad inflation repricing) or twist (term-structure of expectations shifting).

2 hmm_inflation_regime

Gaussian HMM on a feature vector (breakeven changes, real-yield changes, CPI surprises, energy-price changes, rates vol) returning state probabilities. Surfaces regimes such as rising-inflation/tightening, falling-inflation/easing, stagflation, and Goldilocks. (Hamilton 1989)

2 inflation_risk_premium

Decomposes breakevens into expected inflation, inflation risk premium, and liquidity premium components using affine term-structure models (DKW). Output is explicitly model-dependent and labelled accordingly. (D'Amico, Kim and Wei 2018)

5.5 Inflation Swaps

21 zero-coupon inflation swaps, 3 currencies. USD ZCIS (CPI-U), EUR ZCIS (HICPxT), GBP ZCIS (RPI). Seven tenors each (1Y to 30Y).

Zero-coupon inflation swaps (ZCIS) are the OTC derivative analogue of linker-implied breakevens. A ZCIS rate is the par fixed rate against realised inflation; economically it measures inflation compensation, not pure expected inflation. Swaps avoid some bond-specific distortions (no deflation floor, no linker scarcity premium, no on-the-run/off-the-run effects) but introduce collateral, clearing-house, and dealer-balance-sheet effects.

The swap-breakeven basis (ZCIS minus bond-implied breakeven) is a core cross-module RV metric. The 5Y5Y EUR inflation-linked swap rate is the ECB's canonical long-horizon inflation-expectations indicator. The 2008 TIPS liquidity crisis is the textbook example of why both bond and swap inflation measures are needed: bond breakevens collapsed to negative levels for liquidity reasons while swaps remained closer to true compensation, and the basis itself became informative.

Tool Inventory

1A inflation_swap_rate_level

Returns ZCIS rate for any (curve_family, tenor) with period changes, 252d high, low, percentile, and z-score. The inflation analogue of rate_level.

1A inflation_swap_forward

Forward inflation swap rate computed from two ZCIS pillars using the fixed house compounding convention. The 5Y5Y EUR inflation-linked swap rate, the ECB's most-watched anchoring metric, flows through this tool.

1A inflation_swap_curve_spread

Tenor spread on a single ZCIS curve. Surfaces front-end-led versus long-end-led inflation repricing.

1A cross_market_inflation_swap_spread

USD vs EUR vs GBP inflation swap differential at matched tenor or forward window, with index-family caveats applied automatically. CPI-U, HICPxT, and RPI are not fungible inflation measures.

1A swap_breakeven_basis_simple

ZCIS minus bond-implied breakeven at matched tenor. The canonical sign convention. Core RV / liquidity metric and the primary cross-module link to inflation-indexed bonds.

1A inflation_swap_event_change

Pre-event vs post-event repricing of the ZCIS curve. Used for CPI release attribution, central-bank meeting impact, and energy-shock attribution.

1A inflation_swap_scanner

Iterates rate level across all USD/EUR/GBP ZCIS pillars for z-score extremes.

1B seasonality_adjusted_front_end

Decomposes a front-end ZCIS rate into known CPI prints, expected remaining inflation, and a residual risk-premium / technical component. Seasonal model is PM-overridable. Critical for 1Y and 2Y ZCIS interpretation, where index lag and seasonality dominate.

1B inflation_curve_builder

Builds a smooth zero / forward inflation curve from ZCIS pillars under a chosen interpolation method, seasonality treatment, and extrapolation rule.

1B uk_rpi_cpih_transition_monitor

Estimates the pre-2030 vs post-2030 RPI-CPIH wedge priced into the GBP RPI swap curve. Used to monitor pricing of the UK RPI methodology change.

1B swap_breakeven_basis_exact

Production-grade matched-bond basis with carry, deflation-floor, and convention adjustments. The serious version of swap_breakeven_basis_simple, used when the basis trade itself is being considered.

1B inflation_swap_carry_roll

Computes expected carry and roll for a ZCIS position over a horizon, using OIS discount curves, curve roll, index lag, and known/forecast CPI. Cross-domain dependency on OIS curves.

2 pca_inflation_swap_curve

PCA on ZCIS rate changes per curve family. Returns global vs local inflation factors.

2 hmm_inflation_swap_regime

HMM regime model on a feature vector spanning ZCIS rates, breakevens, real yields, CPI surprises, and energy prices. (Hamilton 1989)

2 inflation_risk_premium_swaps

Affine or state-space decomposition of swap rates into expected inflation and inflation risk premium. Output is explicitly model-dependent.

5.6 Policy Futures (STIR)

24 contracts, 3 curve families. Three-Month SOFR Futures (SFR1 to SFR8), Three-Month Euribor Futures (ER1 to ER8), Three-Month SONIA Futures (SFI1 to SFI8).

STIR (short-term interest rate) futures are the highest-frequency, highest-liquidity expression of central bank policy expectations. A SOFR future prices the compounded SOFR rate over its reference quarter; Euribor futures price the 3-month Euribor fixing on the last trading day. Inverse pricing applies: a higher futures price means a lower implied rate, which is dovish.

Per-meeting pricing decomposed from the futures strip is the most-asked front-end analytic alongside OIS meeting pricing, and the futures-vs-OIS basis surfaces arbitrage and hedging dislocations across the two markets.

Tool Inventory

1A **futures_price_level**

Returns last price, implied rate (100 minus price), period changes, percentile, z-score, volume, and open interest for any STIR contract.

1A **futures_implied_rate**

Computes the implied reference rate from price using the inverse-pricing convention. Always displayed alongside price in front-end views.

1A **futures_strip_snapshot**

Returns the first eight contracts in a strip with price, implied rate, volume, open interest, expiry, and strip position. The morning STIR-strip dashboard.

1A **futures_calendar_spread**

Implied-rate spread between any two strip positions. Surfaces front-end curve slope.

1A **futures_butterfly_simple**

Strip curvature in implied-rate space for any three contracts with default 50-50 wing weighting. Used in curve-trade screens.

1A **futures_pack_average_simple**

Simple-average implied rate over a fixed contract group (whites, reds, greens, blues). Used in pack/bundle dashboard views.

1A **futures_cross_market_spread**

Cross-CB implied-rate differential at matched strip positions. Used in cross-central-bank divergence screens (e.g., SOFR strip vs Euribor strip).

1A **futures_scanner**

Scans price moves, implied-rate moves, volume, and OI z-scores across all 24 contracts and returns ranked extremes.

1B **policy_path_from_futures**

Constructs the implied policy path from the futures strip, given the meeting calendar, policy target series, and benchmark-basis assumptions. The futures analogue of OIS per_meeting_pricing.

1B **per_meeting_pricing_from_futures**

Decomposes futures-implied rates into per-meeting hike/cut amounts or probabilities. Exposes step-size, reference-rate-basis, and contract-period-mapping assumptions. Should be cross-checked against the OIS per_meeting_pricing tool.

1B event_repricing_futures

Pre-event vs post-event change by contract, pack, and strip slope. Daily version is close-to-close; intraday version requires intraday futures data (currently aspirational).

1B futures_ois_basis

Compares futures-implied rates with OIS-derived forward rates over matched contract periods. Returns basis level, period changes, z-score, and methodology metadata. Cross-module link to the OIS module.

1B convexity_adjusted_futures_forward

Applies a chosen convexity / margining adjustment to futures-implied forwards. Used when comparing futures to OTC OIS or when constructing a precise forward curve.

1B pack_bundle_analytics

Custom packs and bundles with simple, DV01/tick-value, or user-specified weights. Returns average implied rate, DV01 / tick-value exposure, and P&L per basis point. Distinguishes serial from quarterly contracts before applying whites/reds/greens labels.

2 pca_policy_futures_strip

PCA on implied-rate changes across the strip. Returns level, slope, and curvature factor loadings.

2 hmm_policy_pricing_regime

HMM on strip levels/slopes, OIS forwards, realised volatility, event features, and cross-market spreads. (Hamilton 1989)

2 terminal_rate_distribution

Combines futures, OIS, and (where available) options to estimate a distribution of terminal-rate outcomes. Output is explicitly probabilistic, not a single-point read.

FX Universe

Three instrument domains. Forty-plus tools across three buckets.
Nine G10 spot pairs plus cross-rates · G10 forward curves at seven tenors
G10 implied volatility surfaces (ATM, RR, BF at five tenors).

6.1 FX Spot

9 G10 pairs: EUR/USD, USD/JPY, GBP/USD, USD/CHF, AUD/USD, NZD/USD, USD/CAD, USD/NOK, USD/SEK. Plus derived G10 cross-rates and DXY decomposition.

Spot FX is the most liquid market in the world. The nine G10 pairs in scope are the foundational price for every FX-adjacent workflow: carry trades are measured against spot, forwards are quoted as spot plus points, options are priced from spot, and macro views on monetary policy divergence, terms of trade, and risk appetite are all ultimately expressed through spot.

The USD is base or quote in every G10 pair by convention. Cross-rates (EUR/JPY, GBP/CHF, AUD/NZD, etc.) are derived synthetically. The DXY index is a geometrically weighted basket of six pairs with fixed weights (EUR 57.6%, JPY 13.6%, GBP 11.9%, CAD 9.1%, SEK 4.2%, CHF 3.6%), unchanged since 1973, which makes it a noisy proxy for broad USD strength but the canonical one that every macro PM monitors.

Tool Inventory

1A **fx_spot_level**

Returns spot level, 1d/5d/22d changes (pips and percent), 252d high, low, percentile, and z-score for any G10 pair. The FX analogue of yield_levels. Entry point for every directional workflow.

1A **fx_scanner**

Iterates fx_spot_level across the G10 universe. Returns ranked extremes with z-score sign and magnitude, 1M and 3M momentum percentages, and a diagnostic_flags array of mechanically derived state tags (ZSCORE_EXTREME_HIGH, MOMENTUM_POSITIVE_1M, NEAR_252D_LOW, etc.). Tags describe data state only; no directional verdicts.

1A **cross_rate_panel**

Computes synthetic cross-rate from two USD pairs (e.g., EUR/JPY = EUR/USD x USD/JPY). Returns level, changes, z-score. Surfaces non-USD bilateral dynamics that the USD-pair view obscures.

1A **period_changes_panel**

Computes 1d/5d/22d/63d/252d/YTD changes in pips and percent. Returns structured panel. Building block for the morning FX briefing display.

1A **realised_vol_default**

Close-to-close annualised realised volatility at default 22d window. Returns current vol level, 252d z-score, and time series. Entry point for realised-vs-implied workflows.

1A **dxy_components_decomposition**

Decomposes a DXY index move into per-pair contributions using fixed ICE weights. Surfaces whether a dollar move was broad-based or driven by a single bilateral. Used in morning briefing.

1B **realised_vol_custom**

PM-overrideable window and estimator choice (close-to-close, Parkinson, Garman-Klass, Yang-Zhang). Returns the vol series and multi-estimator comparison. Parkinson/GK capture intraday range missed by close-to-close.

1B **beta_adjusted_pair**

Rolling OLS of one pair on another. Returns beta, residual, residual z-score, hedge ratio. Used in cross-pair RV where raw spreads mislead due to vol differences (e.g., AUD/USD vs NZD/USD).

1B **rolling_correlation**

Rolling pairwise correlations across a pair set. Returns correlation panel and regime indicator from correlation-structure shifts. Used in regime classification.

1B fx_hedged_yield

Composite cross-domain workflow: consumes sovereign yield, OIS rate, forward points, and cross-currency basis from their canonical owners; returns FX-hedged yield pickup decomposed by component. Answers: "What does a JPY investor earn holding USTs after FX hedging?"

1B historical_replay_fx

Replays daily spot path through a position spec. Returns P&L path, drawdown, vol, worst day. The January 2015 CHF crash and March 2020 dollar squeeze are preconfigured windows.

2 pca_g10_fx

PCA on G10 FX returns. Typical PC interpretation: dollar factor, risk factor, regional. Foundational for factor attribution and correlation-regime workflows. (Lustig, Roussanov and Verdelhan 2011)

2 hmm_fx_regime

Gaussian HMM on FX features (DXY changes, realised vol, AUD/JPY as risk proxy). Returns state probabilities, transition matrix, current state. (Hamilton 1989)

2 garch_fx_vol

GARCH(1,1) on pair returns. Returns conditional volatility forecast and vol-of-vol. Used in vol-regime workflows and vol-targeted position sizing. (Bollerslev 1986)

2 cointegration_test_fx

Cointegration test on pairs plus optional macro covariates (e.g., relative CPI for PPP). Returns p-value, equilibrium relationship, residual. Most relevant for PPP-style valuation anchors. (Engle and Granger 1987; Johansen 1991)

6.2 FX Forwards

G10 deliverable forwards and FX swaps. Standard tenor grid: 1W, 1M, 2M, 3M, 6M, 9M, 12M on all nine G10 pairs. Plus cross-rate forwards.

FX forwards are bilateral agreements to exchange currencies at a future date. Economically, forward points reflect the interest-rate differential between the two currencies: the currency with the higher rate trades at a forward discount. FX swaps (spot plus offsetting forward) are the dominant inter-dealer instrument, with \$3.99 trillion per day in the BIS 2025 Triennial Survey (42% of total FX turnover).

For macro PMs, forwards are the vehicle for FX carry trades and the observed side of the covered interest parity (CIP) relationship. The forward-implied yield differential compared to the OIS-implied yield differential reveals the cross-currency basis, a measure of dollar funding stress and swap-market dislocation. Year-end and quarter-end turn effects in forward points are structural calendar artifacts that matter for carry timing.

Tool Inventory

1A forward_level

Returns outright forward, forward points, period changes, 252d z-score for any (pair, tenor). Mirrors `fx_spot_level`. Entry point for forward analytics.

1A forward_scanner

Iterates `forward_level` across G10 pairs and tenors. Returns ranked extremes with `diagnostic_flags` including `FORWARD_POINTS_EXTREME_HIGH`, `CARRY_EXTREME_HIGH`, `CURVE_INVERTED`, `NEAR_YEAR_END_TURN`, and consumed cross-domain `BASIS_EXTREME` flags.

1A implied_yield_differential

Computes $(F/S - 1) \times 365/\text{days}$. Returns the forward-implied interest rate differential, its z-score, and time series. The observed side of the CIP relationship; combined with OIS differentials reveals the cross-currency basis.

1A forward_period_changes

Computes 1d/5d/22d/63d/252d/YTD changes in outright and points. Building block for the forward-market morning display.

1B carry_metric

Computes three carry variants simultaneously: forward premium return (default headline), log forward premium, and unannualised pip P&L. Returns all three with explicit sign convention and base-currency perspective. The sign convention is never implicit.

1B carry_ranked_universe

Iterates `carry_metric` across G10 at a chosen tenor (default 3M). Returns ranked list with carry values, signs, and realised vol per pair for risk-adjusted carry screening.

1B forward_curve_fitter

Fits smooth forward curve from the pillar grid using chosen interpolation (linear, log-linear, monotone cubic spline). Returns fitted curve and per-pillar residuals. Sensitive to year-end turn effects.

1B forward_curve_shape

Curvature metric on forward points at three tenors (default 1M-3M-6M butterfly). Returns value, changes, z-score. Surfaces unusual term-structure curvature in forward markets.

1B forward_curve_roll_yield

Computes expected daily P&L from passage of time alone, holding spot and curve constant. Returns roll yield in pips per day and annualised. Used in carry-trade total-return analysis.

1B beta_adjusted_forward_vs_spot

OLS of forward-points returns on spot returns. Residual z-score is a canonical forward-market dislocation signal: forward-point moves uncorrelated with spot indicate funding or basis pressure.

1B historical_replay_forward

Replays forward and spot path through a carry position. Returns P&L decomposed into spot, forward-points, and basis components. March 2020 dollar funding squeeze is preconfigured.

2 pca_forward_curve

PCA on forward-points changes across the tenor grid. Typical interpretation: PC1 = parallel level shift (rate differential), PC2 = slope (relative cycle expectations), PC3 = curvature.

2 hmm_forward_basis_regime

Gaussian HMM on forward-points and basis features. March 2020 must be in-sample for the funding-stress regime to be identifiable. (Hamilton 1989)

2 garch_forward_vol

GARCH(1,1) on forward-points changes. Returns conditional vol forecast. Used in vol-targeted carry sizing. (Bollerslev 1986)

6.3 FX Options & Vol Surface

G10 implied volatility surfaces. ATM, 25D RR, 25D BF, 10D RR, 10D BF at 1W, 1M, 3M, 6M, 12M on all nine G10 pairs. Plus SABR and Vanna-Volga surface construction.

FX options quote in implied volatility, the input to a Garman-Kohlhagen pricing model that recovers the market's option price. The institutional vol surface is built from three quoted instruments per tenor: ATM volatility (level), 25-delta risk reversal (smile asymmetry: OTM calls vs OTM puts), and 25-delta butterfly (smile convexity: OTM options vs ATM). Together, ATM + RR + BF uniquely define three smile points that can be interpolated into a full surface. (Garman and Kohlhagen 1983; Clark, Foreign Exchange Option Pricing)

Convention sensitivity is a first-class concern. FX options have at least eleven convention fields (delta convention, ATM convention, premium currency, smile interpolation, cut time, and more) that vary by pair, tenor, and source. A 25-delta risk reversal can differ by multiple vol points depending on which convention set is assumed. Every vol surface output in Macro Copilot carries explicit convention metadata so the PM knows exactly what assumptions produced the number.

Tool Inventory

1A **vol_level**

Returns ATM vol, 25D RR, 25D BF mid levels with explicit convention metadata, period changes, 252d z-scores for any (pair, tenor). The vol-surface analogue of `fx_spot_level`. Entry point for vol monitoring.

1A **vol_scanner**

Iterates `vol_level` across G10 pairs and tenors. Returns ranked extremes with diagnostic_flags including `ATM_VOL_EXTREME_HIGH`, `RR_EXTREME_NEGATIVE`, `BF_EXTREME_HIGH`, `VRP_COMPRESSED`, `TERM_STRUCTURE_INVERTED`, `NEAR_CB_EVENT`, and `CONVENTION_DIVERGENT`.

1A **smile_reconstruction**

Reconstructs OTM call and put vols at 25-delta from ATM + RR + BF primitives. Returns smile points with full convention metadata. Building block for surface analysis and convention validation.

1A **gk_pricing**

Garman-Kohlhagen option premium and Greeks (delta, gamma, vega, theta) from spot, strike, rates, vol. Cross-domain dependency on OIS curves for interest rates. Primitive for every pricing and revaluation tool.

1A **vrp_computation**

Computes vol risk premium = implied vol minus realised vol (consumed from `fx_spot` module). Returns time series, current value, 252d distribution. Surfaces compressed VRP (complacency) and elevated VRP (risk aversion).

1A **atm_term_structure_panel**

ATM vol across the full tenor grid (1W through 12M). Returns term structure plus slope metrics (1M-3M, 3M-12M). Surfaces term-structure shape as a vol regime state variable.

1B **event_vol_decomposition**

Decomposes ATM term structure into smooth non-event vol and event-implied vol around an upcoming CB meeting. Returns event vol level, its z-score, and the smooth curve stripped of the event. (Reiswich and Wystup methodology)

1B **sabr_calibration**

Calibrates SABR parameters from ATM + RR + BF (and optionally 10D points). Returns fitted parameters and residuals. Beta fixed at 0.5 by default (FX practitioner standard, PM-overridable). Used for smile extrapolation and rich/cheap identification.

1B vanna_volga_smile

Constructs smile vol at a target strike/delta via the Vanna-Volga replication method. Returns vol with replication-portfolio decomposition. Standard FX practitioner method for cases where SABR is too parametric or three-strike spline is too simple. (Castagna, FX Options and Smile Risk)

1B strike_from_delta

Converts a target delta to a strike under a chosen convention set. Different conventions produce materially different strikes. Convention validation is the primary purpose. Cross-domain dependency on OIS rates and forwards.

1B convention_sensitivity_audit

Recomputes smile points under all alternative convention sets (delta convention swap, ATM convention swap, source screen swap). Returns discrepancy matrix. Surfaces which assumption drives any observed surface anomaly. Rarely automated at funds; first-class tool here.

1B beta_adjusted_vol_pair

OLS of one pair's vol on another's at matched tenor. Returns beta, residual, residual z-score. Used in cross-pair vol RV.

1B historical_replay_vol

Replays vol surface and spot path through an options position. Returns P&L path, drawdown, Greeks evolution. March 2020 vol regime preconfigured.

2 hmm_vol_regime

Gaussian HMM on vol surface features (ATM, RR, BF, term slope). Returns regime probabilities. March 2020 must be in-sample. (Hamilton 1989)

2 pca_vol_surface

PCA on full vol surface panel (ATM x tenor + RR x tenor + BF x tenor). Typical interpretation: PC1 = parallel level, PC2 = term slope, PC3 = smile skew, PC4 = smile convexity.

2 garch_vol_of_vol

GARCH(1,1) on ATM vol changes. Returns conditional vol-of-vol forecast. Used in vol-of-vol-targeted sizing and as a regime indicator. (Bollerslev 1986)

7. Workflows in Action

The product's value is not in individual tools, but in their composition into complete research workflows. Each workflow below is triggered by a single natural-language prompt and executes 2 to 6 deterministic tool calls before returning a synthesised, cited answer.

Morning briefing. *Summarize today's rate moves.* The orchestrator dispatches scanner across the sovereign and OIS universes, identifies extremes (for example, Italy 1Y at +2.91 sigma, Bono 1Y at +3.17 sigma, Bund 1Y at +2.61 sigma), classifies regimes per major curve via `curve_regime`, pulls cross-market context via `cross_market_spread`, and returns a thematically grouped narrative. In Figure 5 above, this exact workflow runs in 23.1 seconds with two tool calls and synthesises a divergent-Europe story across sovereigns and OIS, plus a key-insight summary on the sovereign-OIS basis.

Central bank pricing. *What is priced for the next three FOMC meetings?* `per_meeting_pricing` decomposes the SOFR OIS strip and returns implied changes per meeting (e.g., June minus 4bp, July minus 11bp, September minus 19bp); `forward_rate` adds the 1Y1Y terminal-rate proxy; `policy_path_since_event` compares against the path implied pre-payrolls. The copilot summarises the shift with an explicit risk-neutral caveat.

Cross-asset relative value scan. *Scan for any sovereign yield above 2 sigma and show me cross-market context.* scanner returns the full extreme list; for each candidate, `cross_market_spread` pulls the relevant benchmark spread; `beta_adjusted_spread` checks whether the dislocation survives beta normalisation. Output is a ranked table with commentary: "Italy 2Y at +2.91 sigma raw, but BTP-Bund 2Y spread only at +0.6 sigma after beta. The move is mostly common rates, not Italy-specific."

Inflation regime check. *Where are 5Y5Y breakevens versus history, and how does the swap basis look?* `forward_breakeven_simple` returns the spot 5Y5Y plus z-score; cross-references against the ECB's preferred 5Y5Y EUR ILS via `inflation_swap_forward`; `swap_breakeven_basis_simple` flags whether the bond and swap measures agree. Surfaces both anchoring and basis stress in a single response.

8. Status & Roadmap

BUILT VALIDATED	& Sovereign bonds (6 tools live, all 1A) and OIS swaps (5 tools live, all 1A). Full data pipeline operational: Bloomberg via xbbg into TimescaleDB (instrument_master plus market_data_daily hypertable). LangGraph orchestration with MCP tool servers. UI surfaces functional: Daily Monitor, Workspace, Copilot, Rates page. Validation: 30-prompt gauntlet, 29 of 29 correct routings, 29 of 29 correct tool selections, 59 of 59 correct math.
IN PROGRESS <i>(next 4 to 6 weeks)</i>	Bond futures (CTD identification, basis, DV01), inflation-linked tools (breakevens, seasonal carry), inflation swap tools, policy futures tools (per-meeting pricing, futures-OIS basis). Bucket 1B parameterised tools across all instruments. FX agent under parallel build by cofounder using the same architecture and bucket discipline.
PLANNED <i>(3 to 6 months)</i>	Bucket 2 statistical tools (PCA, HMM, term-premium decomposition). Swaptions and rates volatility module. Cross-currency basis module. Credit agent. Policy / events agent (central bank communication parsing, CPI release workflows). Multi-agent cross-domain orchestration (e.g., FX-hedged yield pickup composing rates and FX agent outputs). Production deployment.

9. Explicit Scope Boundaries

The product is an augmentation tool, not a replacement for PM judgement. The following are deliberately out of scope and will remain so:

- **Trade recommendations.** The product surfaces analytics and context. It does not say buy or sell.
- **Position sizing.** Risk management is the PM's domain. The product provides inputs (DV01, carry, vol) but does not optimise portfolios.
- **Macro forecasting.** The product surfaces market-implied expectations and historical patterns. It does not forecast economic variables.
- **Execution.** No order routing, no broker connectivity, no trade-lifecycle management.
- **Proprietary alpha signals.** Bucket 2 tools provide statistical decompositions, not trading signals. Every output is labelled as model-dependent.